

Calibration and validation strategy for electromechanical cardiac digital twins

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eLife Assessment

This is a potentially **important** study that explores the relevant range of parameter values for calibration and validation of cardiac electromechanics in ventricular models. Although much of the work presented is **solid**, the evidence provided to support the authors' key scientific claims is **incomplete**, especially as it relates to the emphasis on standardized validation and verification approaches. Notably, the level of model personalization presented in this work falls short of the threshold for what could reasonably be called a "digital twin", even by the relatively relaxed standards that have emerged in computational physiology and related fields in recent years.

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Abstract

State-of-the-art cardiac electromechanical modelling and simulation form the basis for recent developments in cardiac Digital Twin technologies. However, a comprehensive evaluation of electromechanical models at cellular, tissue, and organ level has yet to be performed that addresses both ECG and pressure-volume biomarkers. Such an evaluation would build credibility for applications of cardiac Digital Twins in clinical research and therapy development.

We aimed to follow ASME V&V40 standards to develop a strategy for calibration, validation, and uncertainty quantification of ventricular electromechanical Digital Twins under healthy conditions. We performed a multi-scaled review of ventricular electromechanics to compile a dataset for calibration and validation incorporating ECG, pressure-volume, displacement, and strain biomarkers.

When applied to a biventricular multiscale model, we achieved healthy calibrated values for the QRS duration (89 ms), QT interval (360 ms), left ventricular ejection fraction (LVEF) (51 %), peak systolic pressure (14 kPa), end diastolic (110 mL) and end systolic volumes (50 mL), peak ejection flow rate (180 mL/ms). Model validation was performed by comparison to displacement and strain biomarkers including systolic atrioventricular plane displacement

(1.5 cm), systolic fibre strain (-0.18) and longitudinal strain (-0.15). Sensitivity analysis of model parameters at cellular and ventricular scales was also performed. We quantified the effects of variability in ionic conductance, mechanical stiffness, cross-bridge cycling dynamics, and systemic circulation on action potential and active tension dynamics at the cellular scale, and on ECG, pressure-volume, displacement, and strain biomarkers at the ventricular scale. Simulations showed that the relationship between healthy LVEF and T wave biomarkers was primarily underpinned by variability in L-type calcium channel conductance and SERCA activity through multi-scale effects. In this study, we pave the way towards credible cardiac electromechanical Digital Twins by setting the basis for a strategy for calibration and validation based on both ECG and mechanical biomarkers.

Introduction

Digital Twins are playing an increasingly important role in healthcare to enable tailored therapy design for precision medicine. The importance of this emerging technology was attested to by recent regulatory attention and the publication of the ASME V&V40 Standard in 2018¹. The Standard provided a risk-based framework for evaluating their credibility for biomedical applications.

At the core of the cardiac Digital Twin are realistic mechanistic models that can coherently assimilate multi-modal and multi-scale data and can explain disease mechanisms and predict therapy outcomes^{2–4}. An important aspect of cardiac function is the coupling of electrophysiology and mechanics in the ventricles^{5–7}. To describe electromechanical function, Digital Twin models are built on mathematical descriptions of electromechanical coupling at cellular, tissue, and organ scales. The biophysical detail gives the model the advantage of high explainability and predictive power over statistical or machine learning approaches.

Electromechanical cardiac Digital Twins have had broad applicability in cardiology, such as in unravelling the relationship between the electrocardiogram (ECG) and mechanical deformation⁸, the ECG and ejection fraction in post-myocardial infarction⁹, demonstrate how mechanical deformation provides triggers and substrate for arrhythmia¹⁰ and affects re-entrant wave stability¹¹, and predicting therapy outcomes in hypertrophic cardiomyopathy¹² and heart failure¹³. Recent developments have enabled increasingly personalised cardiac Digital Twins through assimilation of omics, imaging, blood pressure, and electrocardiogram (ECG) data^{6,14–16} and promises to deliver patient-specific therapy planning¹⁷ in the near future.

Despite significant developments, a lack of thorough reviews of model credibility considering both electrophysiological and mechanical properties from cell to organ scales hinders wider acceptance of cardiac Digital Twin technologies. Thus far, cardiac applications of the ASME V&V40 standards have focused on device optimisation using fluid-structure interaction models for left ventricular assistive devices¹⁸ and for artificial heart valves¹⁹, but this has yet to be applied to ventricular electromechanical simulations. Examples of rigorous model evaluation at cellular scale^{20–22} highlight the importance of strictly separating calibration and validation data²⁰ and consideration of both electrophysiological and mechanical biomarkers²³. At the ventricular scale, most studies focused either on the ECG⁸ or pressure-volume biomarkers^{24–27,18,28} but not on both, and deformation and strain biomarkers are rarely considered. Simulation studies have investigated how model parameters relate to specific aspects of cardiac electromechanical function, such as the atrioventricular plane displacement²⁸, and how model deformations relate to ECG morphologies⁸, but not in a comprehensive manner. Large scale global sensitivity analyses studies of pressure and volume biomarkers have analysed effects from anatomical variations²⁵ and contractility and haemodynamic parameter variations²⁹, without any analysis of concurrent ECG biomarker effects. Furthermore, the strategies and data used for calibration and validation in these studies were typically not discussed^{5,6,25}.

Therefore, in this study we aim to propose a calibration and validation strategy based on a thorough review of the literature for model parameters and considering both ECG and mechanical biomarkers for evaluating ventricular electromechanical Digital Twins, following ASME V&V40 principles. We apply this to a multiscale, biventricular model with the aim to 1) reproduce healthy pressure-volume, ECG, deformation, and strain dynamics, and 2) investigate multi-scale mechanisms that underpin the relationship between ECG and pressure-volume biomarkers.

Materials and methods

Multi-scale human biventricular electromechanics

Human electrophysiological model

Simulation of the electrical excitation and relaxation through the ventricles was through the monodomain model with orthotropic diffusion along the fibre, sheet, and sheet-normal directions. Ionic current dynamics and calcium handling dynamics were modelled using the human ventricular electrophysiological cell ToR_ORd model²⁰, with extensive validation on experimental action potential morphology and calcium dynamics, as well as responses to multi-channel drug block.

Purkinje-myocardial junctions were modelled using a fast endocardial activation layer with isotropic conduction velocity σ_{endo} on the left and right ventricular endocardial surfaces. Standard 12-lead electrode positions were manually mapped from the heart and torso geometry of a previous study to the current geometry, and the 12-lead ECG was simulated at these electrode positions using the pseudo-ECG method¹⁵.

Mechanical model

The mechanical behaviour was characterised by the balance of linear momentum in a total Lagrangian formulation, considering inertial contributions but ignoring volumetric forces. The reference configuration was the scaled down version of the biventricular mesh (diastasis) $\Omega \subset \mathbb{R}^3$, at $t=0$. The constitutive law for the passive mechanical behaviour of the ventricular tissue was a nearly incompressible version of Holzapfel and Ogden which was orthotropic in mechanical behaviour⁷. The strain energy density function defining this hyper-elastic continuum is:

$$\psi(C) = \frac{K_\alpha}{2} (J-1)^2 + \frac{a}{2b} (e^{b|I_1-3|} - 1) + \sum_{i=f,s} \frac{a_i}{2b_i} (e^{b_i|I_{4i}-1|^2} - 1) + \frac{a_{fs}}{2b_{fs}} (e^{b_{fs}I_{8fs}} - 1)$$

where $J = \det(F)$, $C = F^T F$, bar notation indicates that $\bar{x} = J^{-\frac{2}{3}} x$, f_0 , and indicate fibre and sheet directions, respectively, in the reference configuration such that

$$I_1 = \text{tr} C, I_{4f} = f_0 \cdot C f_0, I_{4s} = s_0 \cdot C s_0, I_{8fs} = \frac{1}{2} (f_0 \cdot C s_0 + s_0 \cdot C f_0), \text{ and } K_{ct} \text{ controls the}$$

compressibility of the material. The reader is directed elsewhere⁷ for the precise form of the passive contribution to the second Piola-Kirchhoff stress tensor.

Boundaries were defined as: epicardium Γ_{epi} , left ventricular endocardium Γ_{LVendo} , right ventricular endocardium Γ_{RVendo} , and the non-endocardial surfaces of the valvular plugs Γ_{valve} ,

$$FS n_0 - J P_{LVendo} F^{-T} n_0 = 0, \text{ on } \Gamma_{LVendo} \times [0, T],$$

$$FS n_0 - J P_{RVendo} F^{-T} n_0 = 0, \text{ on } \Gamma_{RVendo} \times [0, T],$$

$$FS n_0 - (K_{epi} (uv \cdot c_l) u \cdot n_0) n_0 = 0, \text{ on } \Gamma_{epi} \times$$

where n_0 was the outward normal defined on the whole material boundary $\partial\Omega_0$, and K_{epi} (uv c_l) was the stiffness corresponding to an elastic spring boundary condition applied to the endocardium and was described as a step function of the longitudinal coordinate l with magnitude of K_{epi} at values of $0 < uv$ $c_l < 0.85$ (i.e. applied for 85% of the longitudinal dimension of the heart from the apex). This was a simplified version of the method²⁸, which uses an exponential decay formulation at the ‘edge’ of the pericardial constraint rather than a step function. P_{LVendo} and P_{RVendo} were the pressures applied to the left and right endocardial surfaces, respectively, and was prescribed by two independent piece-wise functions that describe the five-phases (not including initiation) of the cardiac cycle:

- *Phase 0: Initiation.* Endocardial pressures were linearly increased to reach diastasis values P_{RVDS} and P_{LVDS} .
- *Phase 1: Active diastolic filling.* Endocardial pressures were linearly increased over time t EDP to end diastolic values EDP_{LV} and EDP_{RV} to mimic the effect of atrial contraction.
- *Phase 2: Isovolumetric contraction.* Electrical activation ensues and active contraction develops. During this, the ventricular pressure was controlled such that the volume was maintained to be constant via:

$$\begin{aligned} dP_{LVendo} &= \frac{-1}{C_{P_{LV}}} dV_{LVendo} - \frac{1}{C_{V_{LV}}} \frac{dV_{LVendo}}{dt} \\ dP_{RVendo} &= \frac{-1}{C_{P_{RV}}} dV_{RVendo} - \frac{1}{C_{V_{RV}}} \frac{dV_{RVendo}}{dt} \end{aligned}$$

Where $C_{P_{LV}}$, $C_{P_{RV}}$, $C_{V_{LV}}$, $C_{V_{RV}}$, were the inverse of penalty terms for volume difference and volume rates for the left and right ventricles, respectively. The volume rate term was used as stabilisation against spurious oscillations of the ventricular pressure, which might occur due to inertial effects.

- *Phase 3: Ejection.* When the ventricular pressure surpasses the aortic P_{AO} and pulmonary artery P_{PA} pressures, the ejection phase begins. To model the blood pressure of the systemic circulation system during the ejection phase, the following two element Windkessel model was used:

$$\begin{aligned} C_{AO} \frac{dP_{AO}}{dt} + \frac{P_{AO}}{R_{AO}} &= -\frac{dV_{LVendo}}{dt} \\ C_{PA} \frac{dP_{PA}}{dt} + \frac{P_{PA}}{R_{PA}} &= -\frac{dV_{RVendo}}{dt} \end{aligned}$$

Where C_{AO} and C_{PA} were the compliance of the aortic and pulmonary arteries, respectively, and R_{AO} and R_{PA}

$$dP_{LVendo} = \frac{-1}{C_{pLAV}} (V_{LVendo} - DSV_{LV}) - \frac{1}{C_{vLAV}} \frac{dV_{LVendo}}{dt},$$

$$dP_{RVendo} = \frac{-1}{C_{pRAV}} (V_{RVendo} - DSV_{RV}) - \frac{1}{C_{vRAV}} \frac{dV_{RVendo}}{dt}$$

were the resistance of the aortic and pulmonary circuits, respectively. During the ejection phase, the ventricular pressures P_{LVendo} and P_{RVendo} were modelled as equal to the arterial pressures P_{AO} and P_{PA} respectively, disregarding negligible pressure gradients across the arterial valves.

- *Phase 4: Isovolumetric relaxation.* This phase was triggered when the ventricular flow reverses $V_{LVendo} > 0$, and $V_{RVendo} > 0$, and it follows the same formulation as the isovolumetric contraction Phase 2.
- *Phase 5: Passive filling.* This phase begins when the endocardial pressure drops below the left and right atrial pressures P_{LA} , P_{RA} . The pressure was prescribed so that the two ventricular volumes were returned to its diastasis value (DSV_{LV} , DSV_{RV}).

Where C_{pLAV} , C_{pRAV} , C_{vLAV} , C_{vRAV} were the inverse of penalty terms for volume different to DSVs and volume rates, for the left and right ventricles, respectively.

The two ventricular volumes V_{LVendo} and V_{RVendo} were calculated and updated throughout the cardiac cycle by using the divergence theorem and the assumption that the volume was constrained by a flat lid located on the aortic and pulmonary valvular planes, for right and left ventricles respectively, leading to:

$$V_{LVendo} = \int_{\partial\Omega_{LVendo}} J(x \cdot e_{AO}) (e_{AO} \cdot F^{-T} n_0) d\partial\Omega_0,$$

$$V_{RVendo} = \int_{\partial\Omega_{RVendo}} J(x \cdot e_{PA}) (e_{PA} \cdot F^{-T} n_0) d\partial\Omega_0,$$

Where e_{AO} and e_{PA} were defined so that $e_{AO} \cdot n_0 = 0$ on the aortic valve plane, and $e_{PA} \cdot n_0 = 0$ on the pulmonary valve plane.

Electromechanical coupling

Electrophysiological activation begins at Phase 2 of the cardiac cycle and drives the active tension generation. This is modelled at the cellular level by coupling of the Land active force generation²¹ with intracellular calcium dynamics from the human cellular electrophysiology ToRORd model²⁰, henceforth referred to as the human cellular electromechanical ToRORd-Land model. In order to couple the ToRORd to the Land model, the Hill coefficient of calcium cooperativity and tropomyosin rate constant of the Land model were re-fitted in a previous study²³ so that physiological active tension could be produced in response to the calcium transient dynamics of the ToRORd model, which were different to the calcium transient to which the original Land model was fitted.

In the Land model, active force generation is modelled using four cross-bridge cycling states (blocked, unbound, pre-power stroke and the force-generating state) and the transition rates between each were calibrated to human skinned myocyte data²¹. The ToRORd-Land coupling is bidirectional, where calcium from the ToRORd binds to troponin C and drives the transition between the blocked and unbound states of the crossbridge system, while the binding of calcium from the Land model acts as a buffer for the ToRORd calcium transient. Furthermore, the length-dependence of troponin calcium affinity/sensitivity in the Land model means that calcium buffering in the ToRORd is dynamically affected by tissue deformation. The Land model was first constructed using skinned myocyte data, then several of the parameters were re-fitted to enable sufficient ejection at the ventricular scale, including the transition rate from the pre-power stroke to force-generating state of the cross bridge (k_{ws}), the sensitivity of troponin to calcium binding (Ca150)²¹. At the tissue level, the active tension was then added to the stress tensor to drive mechanical contraction⁷.

Literature-informed parameter initialisation

A review of the literature produced a list of initial model parameter values as compiled in **Table 1**, including also reported parameter variability. Passive mechanical properties were extracted from the non-viscoelastic version of the model fitted to human ex vivo stress-strain measurements³⁰. Ionic time constants and conductance values were set to baseline values of the human membrane kinetics ToR_ORd model, which had been calibrated to patch clamp, action potential and calcium transient data²⁰. The active tension Land model parameters were as previous calibrated to achieve a physiological calcium transient²³.

Model assessment strategy

Following examples of the ASME V&V 40-based analysis applied in the context of cardiovascular science^{18,48}, we devised a calibration and validation strategy for human ventricular electromechanical modelling and simulation frameworks by considering the following:

Group	Parameters	Initial value	Literature ranges
Passive mechanics parameters	a, af, as, afs	0.61 kPa, 1.56 kPa, 0.70 kPa, 0.46 kPa	0.61, 1.56, 0.70, 0.46 (fitted to human data shear and axial data) ³⁰ 0.057, 21.503, 6.841, - (fitted to partial shear pig data) ³¹ 0.059, 18.472, 2.481, 0.216 (fitted to full shear pig data) ³¹ 2.28, 1.685, -, - (fitted to biaxial dog data) ³¹ 0.333, 18.535, 2.564, 0.417 ³²
	b, bf, bs, bfs	7.5, 35.31, 33.24, 5.09	7.5, 35.31, 33.24, 5.09 (fitted to human data shear and axial data) ³⁰ 8.094, 15.819, 6.959, - (fitted to partial shear pig data) ³¹ 8.023, 16.026, 11.12, 11.436 (fitted to full shear pig data) ³¹ ; 9.726, 15.779, -, - (fitted to biaxial dog data) ³¹ 9.242, 15.972, 10.446, 11.602 ³²
	Kct	5000 kPa	3333 ³² , 10 - 1000 ³³
Active tension parameters	Tref	120 kPa * 10	120 (fitted to human contraction data) ²¹
	Cross-fibre contraction	30% of Tref	0 – 60% ⁷ , 46% (rabbit heart tissue measurements) ³⁴
Pressure volume control parameters	R_LV	10 kPa ms/mL	75 – 750 mmHg ms/mL ⁷ , 750 mmHg ms/mL ³² , 852 mmHg ms/mL ²¹
	R_RV	33.3 kPa ms/mL	A third of left ventricular value ²¹ , 2.0 +- 0.8 mmHg min/L (control patient measurements) ³⁵
	C_LV	1.5 mL/kPa	0.2 mL/ mmHg ³² , 2.73 mL/mmHg ²¹

Table 1

Initial parameter values and literature basis for the choice of uncertainty ranges for the model inputs. Unless stated explicitly, quoted values from the literature have come from previous modelling choices.

	C_RV	4.5 mL/kPa	Threefold of left ventricular value ²¹
	P ejection LV	10 kPa	73 -78 mmHg (human measurements) ³⁶ , 9 kPa ²¹
	P ejection RV	2.3 kPa	17.5 +- 7.5 mmHg (control patient measurements) ³⁵ , 3 kPa ²¹
	P diastolic cut-off LV	1.3 kPa	10 [7-13] mmHg (control patient measurements) ³⁷ , 0.1 kPa ²¹
	P diastolic cut-off RV	1.5 kPa	14.4 ± 5.8 mmHg, pulmonary diastolic pressure ³⁵
	Atrial filling duration LV	150 ms	~150 ms at resting cycle length 1000 ms (human exercise measurements) ³⁸
	EDP_LV	1.3 kPa	10 ± 1 mmHg (control patient measurements) ³⁹ , ~7.5 mmHg(control patient measurements) ⁴⁰ , 7.5 mmHg ²¹ .
	EDP_RV	1.0 kPa	11.0 ± 6.3 mmHg ³⁵ , A third of left ventricular value ²¹
Pericardial constraint	K_epi	500 kPa/cm	20 kPa/cm with 5e-2 kPa*s/cm for viscous term ⁴¹ , 500 kPa/cm ²⁸ , 0 – 100 kPa/cm (20 kPa/cm specifically for baseline) ⁷
Microstructure	Transmural fibre angle endo to epi	-60 to 60 in LV, 90 to -25 in RV, 0 to 90 at outflow tract	-60 to 60 in LV, 90 to -25 in RV, 0 to 90 at outflow tract ⁴²
	Transmural sheet angle to radial axis	0 degrees	-45 endocardium to 45 epicardium (canine measurements) ⁴³ , 45 endocardium to 0 epicardium (canine measurements) ⁴⁴ , ~45 degrees at diastole, ~0 degrees at systole (swine DTI measurements) ⁴⁵
Heart rate	Cycle length	1000 ms	63+-2 beats/min ⁴⁶ , 72 +- 13 beats/min ⁴⁷

Table 1 (continued)

Question of interest

Can the modelling and simulation framework produce healthy physiological electromechanical function at cellular, tissue, and organ levels including both ECG and pressure-volume loop biomarkers?

Context of use

Human ventricular electromechanics models provide mechanistic explanations for various disease conditions and evaluate therapeutic interventions. Here we focused on model evaluation under healthy conditions to build the foundations of model credibility for disease and therapy investigations in the future.

Quantities of interest

To ensure model relevance to its context of use, the quantities of interest were chosen to be biomarkers with known links to key cardiac diseases, including Torsade de Pointes, myocardial infarction, hypertrophic cardiomyopathy, dilated cardiomyopathy, heart failure, and pulmonary hypertension. Therefore we selected the following biomarkers from the 12-lead ECG (QRS duration, QT dispersion, T wave amplitude, T peak to T end duration, and JT interval), the pressure-volume relationship (left and right end diastolic and end systolic volumes and end systolic pressures, and left ejection fraction, left pressure upstroke velocity, left peak ejection and peak filling flow rate), and from measurements of displacement (end diastolic and end systolic wall thicknesses, atrioventricular plane displacements, systolic percentage volume change, and apical displacement) and strain (systolic global longitudinal strain, radial strain, circumferential strain, and fibre stretch ratio). The relevance of each biomarker to cardiac diseases was summarised in Table 3.

Cellular biomarkers including action potential duration (APD90), active tension peak ($T_{a,max}$), active tension duration (T_{aD50}) and peak upstroke of active tension (dT_a/dt_{max}) were not explicitly included in the biomarker list since they have already been used to calibrate and validation the cellular electromechanical model previously^{20,23}. However, these properties were evaluated in uncertainty quantification (see below) to inform ventricular simulations.

Compilation of biomarker data

The healthy ranges for each quantity of interest (biomarker) were compiled from the literature. Reports from larger datasets were preferred for better estimates of population variability. For this reason, values were preferentially included from the UK Biobank, which is a large-scale biomedical database with health information from half a million UK participants, with more than 40,000 cardiac magnetic resonance recordings available^{49–51}. This is then supplemented by smaller studies as needed. The use of gold standard magnetic resonance imaging data was preferentially included over other imaging modalities such as echocardiography for volumetric and deformation biomarkers.

Separation of calibration vs validation data

The quantities of interest were separated into a calibration and a validation dataset²⁰. The calibration dataset consisted of ECG and pressure-volume biomarkers that clinically defined healthy electromechanical function, while the validation dataset included displacement and strain biomarkers that explored the model's capabilities for providing mechanistic explanations.

Sensitivity analyses strategy

The model was further validated through sensitivity analyses. To showcase versatility, a wider set of parameters were included in the analyses than has explicitly been linked to any disease or therapeutic target.

The following model parameters were included from the cellular electrophysiology (conductances of the fast sodium current G_{Na} , late sodium current G_{NaL} , L-type calcium current G_{CaL} , transient outward potassium current G_{to} , rapid delayed rectifier potassium current G_{Kr} , slow delayed rectifier potassium current G_{Ks} , sodium calcium exchanger G_{NCX} , inward rectifier current G_{K1} , sodium-potassium pump current G_{NaK} , peak calcium release J_{rel} , peak calcium re-uptake $SERCA$, and magnitude of the stimulus current I_{stim}), the cellular active tension generation (peak active tension T_{ref} , calcium sensitivity Ca_{50} , cross bridge transition rates from unbound to pre-powerstroke k_{uw} and from pre- to post-powerstroke k_{ws}), passive mechanical properties (bulk stiffness a , fibre stiffness a_f , sheet stiffness a_s , fibre-sheet shear stiffness a_{fs} , compressibility coefficient Kct , pericardial constraint stiffness K_{epi}), and circulatory haemodynamics (aortic resistance R_{LV} , aortic compliance C_{LV} , aortic ejection threshold pressure $P_{ejectionLV}$).

Model parameter ranges were informed by variabilities in literature reports (summarised in [Table 2](#)). Cellular conductances were uniformly varied from 50 to 200% to cover the experimentally measured ranges as done in previous studies²². For passive mechanics, only the parameters carrying units of stiffness were included in the exploration and their ranges come directly from the literature. The compressibility parameter (Kct), pericardial constraint stiffness (K_{epi}), and the active tension coefficient (T_{ref}) parameters were allowed greater ranges of variation since they do not correspond to experimental measurements and their physical meanings were model-dependent. Similarly, the resistance and compliance parameters and the ejection threshold pressure were allowed a large uncertainty range since they were part of a lumped parameter model and do not directly map to experimental measurements.

A specific example of model calibration, validation and sensitivity analysis

Input data

To demonstrate our strategy we selected a patient-specific (female, age 49, weight 90 kg) biventricular tetrahedral mesh with valvular plugs from four-chamber meshes available from a published database²⁵. Since the mesh was constructed based on imaging data at end diastole, the mesh was scaled down isotropically to achieve a diastasis volume of 80 mL³⁹ to mimic the process of unloading to diastasis. Fibre f_0 , sheet s_0 , and sheet-normal n_0 vectors at resting were generated using a rule-based method designed for biventricular geometry with outflow tracts⁴². Ventricular coordinates were taken from the published database²⁵, and consisted of the longitudinal coordinate ($uv\ c_l$) where $uv\ c_l=1$ at the apex and 0 at the base, a transmural coordinate ($uv\ c_r$), where endocardium was $uv\ c_r=1$ and epicardium was $uv\ c_r=0$, and a rotational coordinate ($uv\ c_\theta$), where $uv\ c_\theta=0$ was in the middle of the left ventricular lateral wall.

Calibration to ECG and pressure-volume biomarkers

The 12-lead ECG signals of a female healthy volunteer with similar myocardial mass (108 g) as the biventricular mesh (96 g) was extracted and beat-averaged¹⁵ for calibration of electrophysiological activation and repolarisation characteristics. The conduction velocities along the fibre and sheet-normal directions were extracted from surgical measurements⁵². The root nodes of earliest activation on the endocardial surface as well as the conduction velocities on the

Mechanical and haemodynamic parameters	Literature ranges	Variability ranges
a, af, as, afs (kPa)	0.61, 1.56, 0.70, 0.46 (fitted to human data shear and axial data) ³⁰ 0.057, 21.503, 6.841, - (fitted to partial shear pig data) ³¹ 0.059, 18.472, 2.481, 0.216 (fitted to full shear pig data) ³¹ 2.28, 1.685, -, - (fitted to biaxial dog data) ³¹ 0.333, 18.535, 2.564, 0.417 ³²	[0.057, 0.61] [1.56, 2.15] [0.7, 6.8410] [0.216, 0.46]
Kct (kPa)	3333 ³² , 10 - 1000 ³³	[10, 500]
Tref (kPa)	120 (fitted to human contraction data) ²¹	[1200, 2400]
Arterial resistance	75 – 750 mmHg ms/mL ⁷ , 750 mmHg ms /mL ³² , 852 mmHg ms/mL ²¹	[500, 999]
Arterial compliance	0.2 mL/ mmHg ³² , 2.73 mL/mmHg ²¹	[0.1, 0.2]
P ejection LV (kPa)	73 -78 mmHg (human measurements) ³⁶ , 9 kPa ²¹	[5, 10]
K_epi (kPa)	20 kPa/cm with 5e-2 kPa*s/cm for viscous term ⁴¹ , 500 kPa/cm ²⁸ , 0 – 100 kPa/cm (20 kPa/cm specifically for baseline) ⁷	[20, 500]

Table 2

Variability ranges for mechanical and haemodynamic parameters explored in ventricular sensitivity analysis.

endocardial surface and transmurally were calibrated using the QRS segment of the ECG signal, while spatial heterogeneity in the slow rectifier potassium current (IKs) was calibrated to the ST-T segment of the ECG signal, using a Bayesian-based inference method^{15,53}. The calibrated model had a transmural conduction velocity of 48 cm/s, and the fibre and sheet-normal conduction velocities were set to 67 cm/s and 44 cm/s as measured using plunge needle in control subjects⁵².

When the initialised parameter values were used in a preliminary electromechanical simulation, the results undershot healthy values of left ventricular peak systolic pressure by 2 kPa, ejection fraction by 32%, stroke volume by 29 mL, and peak ejection rate by 142 mL/ms, and peak filling rate by 207 mL/ms, and overshoot the end systolic volume by 18 mL, and the peak pressure upstroke velocity by 70 kPa/ms. This demonstrated the need for model calibration and explorations of model uncertainties.

Due to the high computational cost of ventricular electromechanics simulations, calibration methods that require high volume of evaluations were infeasible. In this study, we designed a sequential calibration strategy based on knowledge of model sensitivities (described in detail in Appendix 1). This involves a series of sampling and parameter selections as follows, where single parameters were uniformly sampled and multiple parameters were sampled using Latin Hypercube Sampling:

1. Sample ejection pressure threshold, Kct, Cal50 and k_{ws} and select parameter set that gives highest LVEF.
2. Sample arterial resistance R_{LV} and select value that gives best peak systolic pressure.
3. Sample GCaL and select value that gives highest LVEF.
4. Sample k_{ws} and select value that gives best peak ejection rate, dP/dt_{max} and LVEF.
5. Sample diastolic volume change parameter and select value that gives best peak filling rate.

Validation by comparison to deformation and strain biomarkers

The calibrated model was evaluated against displacement and strain biomarkers as listed in Table 3 and not used for model calibration. Atrioventricular plane displacement was calculated by tracking the longitudinal displacement of the top 10% of the geometry using the apex-to-base ventricular coordinate. Wall thickness was calculated using perpendicular projections of the endocardial and epicardial surfaces. Myocardial strains were evaluated at a mid-ventricular short axis slice and at a four-chamber longitudinal, to match DENSE measurements.

Circumferential, radial, and fibre strains were evaluated using the short axis slice, while the longitudinal strain was evaluated using the long axis slice. The strain transients were then time-shifted to begin at the end of diastolic filling (Phase 1) and offset by the end diastolic strain, to match DENSE measurements.

Sensitivity analysis at cellular and ventricular scales

A global sensitivity analysis was first performed at the cellular scale using the human cellular electromechanical ToR_Ord-Land model²³ to identify the key parameters that affect active tension and action potential duration. The conductances of all ionic currents as well as troponin calcium sensitivity and the cross-bridge power stroke rate and myosin head attachment rates (k_{ws} and k_{uw}) were varied from 50% to 200%. The key parameters that affect active tension dynamics and action potential duration were identified through ranking the parameters using the total Sobol index. This analysis showed that the key parameters which affected active tension amplitude, duration, and upstroke were GCaL, SERCA, calcium sensitivity Cal50, the cross-bridge cycling rate (k_{ws}) and GNaL (Appendix Figure 1).

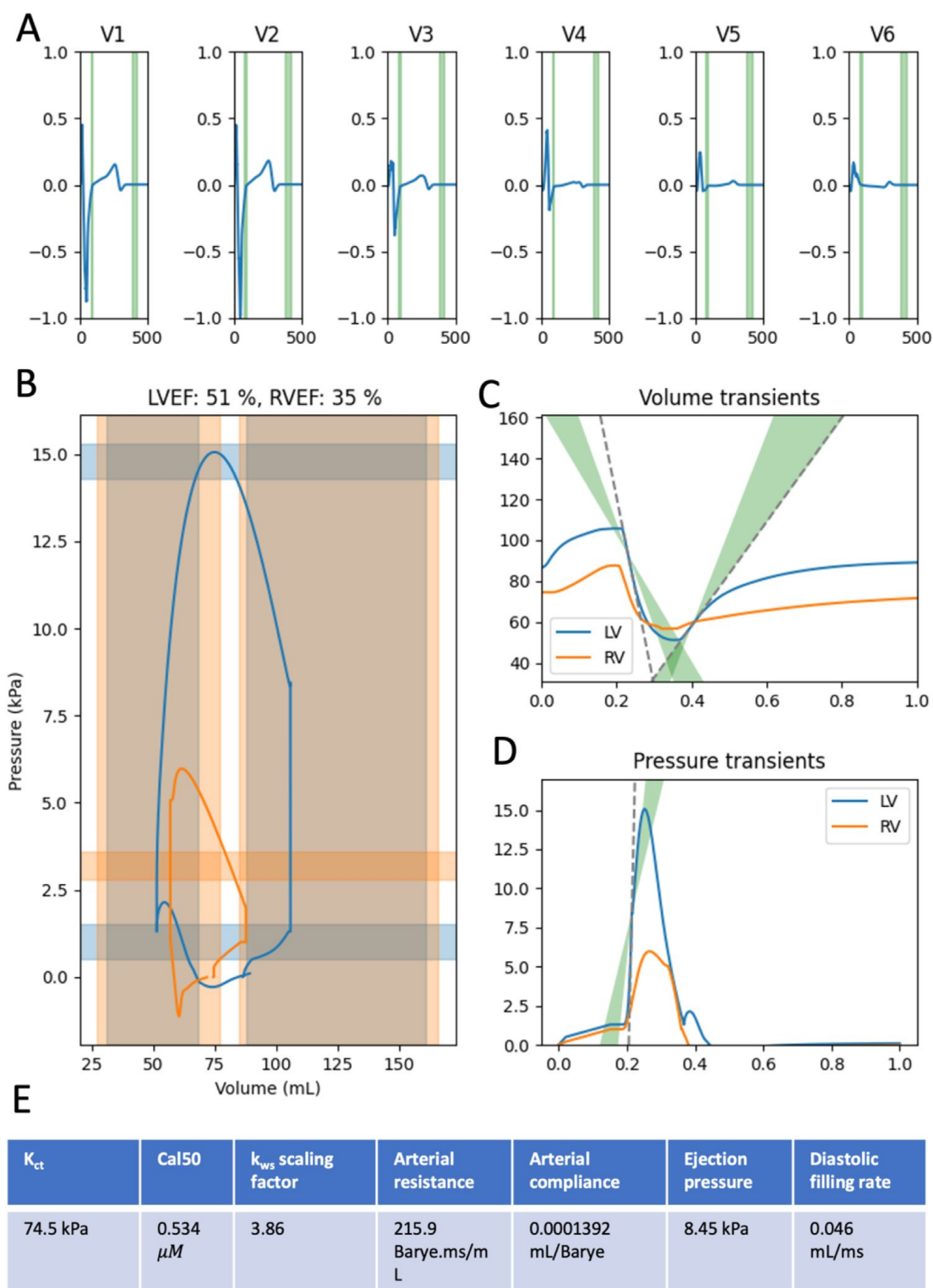


Figure 1

Comparison of (A) simulated ECG, (B) pressure-volume, (C) flow rate, and (D) pressure upstroke biomarkers of the calibrated baseline electromechanical model with healthy population data ranges shown in green, except for (B) where the left ventricular ranges were shown in blue and the right ventricular ranges in orange.

The calibrated model parameters were shown in the table (E).

At the ventricular scale, the mechanical and haemodynamic parameters listed in **Table 2** along with the cellular parameters GCaL, SERCA, Cal50, kws, and GNaL were included one-at-a-time sensitivity analyses, varying each parameter uniformly in a total of eight simulations each over the ranges as specified in **Table 2** for mechanical and haemodynamic parameters and over 50% and 200% for cellular parameters. Each biomarker as listed in Table 3 were evaluated, and linear regression was performed.

Parameter-biomarker relationships that had p-values less than 0.05 and absolute r-values larger than 0.6 were considered significant and linear relationships. The magnitudes of biomarker effects were normalised by the maximum magnitude for each biomarker. To visualise the relative importance of model parameter on simulated biomarker, a summary diagram was generated with lines linking each parameter to each biomarker. The colour of each line indicated a positive (red) or a negative (blue) relationship. The thickness of each line was scaled by the normalised magnitude of the effect. The transparency of each line was scaled by the r-value as measure of the linearity of relationship (Results **Figure 3**).

Computations and software

Cellular electrophysiological simulations and Latin Hypercube Sampling were performed using bespoke MATLAB codes. Coupled cellular electromechanics, as well as biventricular electromechanics simulations, were performed using the high-performance numerical software, Alya for complex coupled multi-physics and multi-scale problems⁵⁴ on the JURECA pre-exascale module supercomputer operated by Juelich Supercomputing Centre, Germany, through a PRACE-ICEI (project, as well as on the ARCHER2 supercomputer provided by the UK National Supercomputing Service through the CompBioMedX project (Computational Biomedicine at the Exascale) funded by EPSRC under grant agreement EP/X019446/1. Code for the generation of simulation files for multi-scale sensitivity analysis, calibration and validation, as well as scripts for evaluation of ECG, PV, deformation and strain biomarkers from ventricular simulations can be found at: https://github.com/jennyhylanwe/Alya_input_setup/

Results

Experimental and clinical dataset for model calibration and validation

Table 3 presents the experimental and clinical dataset for healthy human ventricular electromechanics. For pressure-volume biomarkers and ECG biomarkers, reported values from the UK Biobank¹ were used, where the number of samples reaches the tens of thousands. Catheter measurements of intraventricular pressure dynamics were rare in the literature and have small dataset sizes. Mechanical displacement and strain biomarkers came from MRI studies with specialised sequences (e.g. DENSE strain imaging).

Biventricular electromechanics model calibration to ECG and pressure-volume biomarkers

Following model calibration, the simulated ECG showed good R wave progression from V1 to V6 and QRS duration that fell within the ranges shown in **Table 1** (**Figure 1A**) as well as simulated left ventricular ejection fraction of 51% and peak systolic pressure of 14 kPa, with end diastolic and end systolic volumes of 110 mL and 50 mL (**Figure 1B**), and a peak filling rate of 310 mL/ms (**Figure 1C**), all falling within 95% interval of physiological biomarker ranges as listed in **Table 1**. The right ventricular end diastolic and end systolic volumes also fell within physiological ranges, though the right ventricular ejection fraction did not, as this was not a key

Group	Biomarker	Ranges	Data info	Implication for cardiac diseases
Calibration dataset				
12-lead ECG	JT interval (ms)	313 [294, 335] _{IQR} ⁵⁵	mixed sex, UKBB CMR, n=76,266	In patients with QRS > 120 ms, risk of Torsade de Pointes ^{56,57}
	Tpe (ms)	62 (12) _{IQR} ⁵⁸	mixed sex, UKBB without life threatening arrhythmias, n=51,574	Risk of SCD ⁵⁹
	T wave amplitude (Tpeak) (mV)	0.15 [0.11, 0.19] _{IQR} ⁴	mixed sex, UKBB normal ventricular mass, n=36,956	Risk of SCD in HCM ⁶⁰
	QT dispersion (ms)	56.6 [37.0, 84.0] _{IQR} ⁴	mixed sex, UKBB normal ventricular mass, n=36,956	Severity of CAD ⁶¹
	QRS duration (ms)	89 [81, 97] _{IQR} ⁴	mixed sex, UKBB normal ventricular mass, n=36,956	Poor prognosis in DCM ⁶² and risk stratification for HFpEF ⁶²
Pressure volume	LVEDV (mL)	[88, 161] _{95%} ⁴⁹	female UKBB CMR, n=804	Predictor of exercise capacity in HFpEF ⁶³
	LVESV (mL)	[31, 68] _{95%} ⁴⁹	female UKBB CMR, n=804	Predictor of HFpEF ⁶⁴
	RVEDV (mL)	[85, 166] _{95%} ⁴⁹	female UKBB CMR, n=804	Predictor of functional class in PAH ⁶⁵
	RVESV (mL)	[27, 77] _{95%} ⁴⁹	female UKBB CMR, n=804	Stratifies PAH ^{66,67}
	SVL (mL)	[49, 100] _{95%} ⁴⁹	female UKBB CMR, n=804	Incident rate of HF ⁴⁴
	LVESP (kPa)	16.7 ± 2.2 _{std} ⁶⁸	mixed sex, UKBB,	Associated with

Table 3

Experimental and clinical datasets for calibration and validation of healthy ventricular electromechanical models, with summary of why each biomarker was important due to their implications in cardiac diseases.

Data were presented in the forms: [x, y]_{95%} were 95% confidence intervals, $x \pm y_{std}$ were mean and standard deviations, $x \pm y_{sem}$ were mean and standard error of the mean, [x, y]_{range} were the minimum (x) and maximum (y), x [y, z]_{IQR} were median (x) and first (y) and third (z) quartiles, x (y)_{IQR} were median (x) and interquartile range (y). UKBB indicates values extracted from the UK Biobank. Unit conversions from original source has been done where appropriate to preserve consistency across the entire table. SCD: sudden cardiac death, HCM: hypertrophic cardiomyopathy, DCM: dilated cardiomyopathy, CAD: coronary artery disease, HF: heart failure, HFpEF: HF with preserved ejection fraction, PAH: pulmonary artery hypertension, MI: myocardial infarction, HHD: hypertensive heart disease.

			n=150	hypertrophy ^{68,69}
	RVESP (kPa)	$3.2 \pm 0.4_{\text{std}}$ ⁷⁰	male, radionuclide ventriculography, n=12	Stratifies PAH ^{66,67}
	LVEF (%)	$[51, 70]_{95\%}$ ⁴⁹	female UKBB CMR, n=804	Stratifies HF, MI, and SCD risk ⁷¹
	dP/dt _{max} (kPa/s)	$150 \pm 30_{\text{std}}$ ⁷²	catheter measurement in successful CRT, n=17	Surrogate measure of ventricular contractility in HF ⁷³
	Peak ejection rate (PER) (mL/s)	$410 \pm 102_{\text{std}}$ ⁷⁴	mixed sex, 3D echocardiography, n=15	Valvular regurgitation ⁷⁵ and peak systolic velocity associated with CHD ⁷⁶
	Peak early filling rate (PFR) (mL/s)	$361 \pm 110_{\text{std}}$ ⁷⁴	mixed sex, 3D echocardiography, n=15	Points to relaxation dysfunction in CHF ⁷⁷
Validation dataset				
Displacements and strains	ED wall thickness (cm)	$[0.57, 1.38]_{\text{range}}$ ⁷⁸	mixed sex, range of mean values over 17 AHA regions, UKBB CMR, n=4620	Increased in HCM or HHD ^{79,80}
	ES wall thickness (cm)	$[1.07, 1.749]_{\text{range}}$ ⁷⁸	mixed sex, range of mean values of 17 AHA regions, UKBB CMR, n=4620	Fractional thickening reduced in HCM ⁸⁰
	Atrioventricular plane displacement (AVPD) (ES – ED) (cm)	$[1.4, 1.9]_{\text{range}}$ ⁴⁶	mixed sex, CMR, healthy subjects n=12	Reduced in DCM ⁴⁶
	Systolic percentage volume loss	$13 \pm 4\%$ _{std}	Mixed sex, CMR, healthy subjects, n=115	Less compressibility in HFrEF ⁸¹
	Apical displacement (AD) (ES – ED) (cm)	$[-0.1, 5.1]_{\text{range}}$ ⁴⁶	mixed sex, CMR, n=12, with positive meaning apex shifted towards the base during systole	Predicts CRT outcomes in ischaemic cardiomyopathy ⁸²
	Systolic global longitudinal strain	$-0.16 [-0.16, -0.16]_{\text{IQR}}$ ⁴⁷	four chamber slice, mixed sex, cDTI +	Major predictor of LV dysfunction and HF,

Table 3 (continued)

	(EII)		DENSE, n=30	greater accuracy than LVEF ^{83,84}
	Systolic radial strain (Err)	0.27 [0.22, 0.29] _{IQR} ⁴⁷	Short axis slices, mixed sex, cDTI + DENSE, n=30	Associated with chronic HF ⁸⁵
	Systolic circumferential strain (Ecc)	-0.10 [-0.14, -0.08] _{IQR} ⁴⁷	Short axis slices, mixed sex, cDTI + DENSE, n=30	Associated with chronic HF ⁸⁵

Table 3 (continued)

target for calibration. The calibrated model overshoot the peak pressure upstroke by 30 kPa/ms and peak ejection rate by 180 mL/ms (**Figure 1D**). However, the importance of matching to these two biomarkers is lower compared with other haemodynamic markers due to the relative paucity of reported data: $n=17$ for peak pressure upstroke and $n=15$ for peak ejection rate, compared with $n=804$ for ejection fraction and $n=150$ for peak systolic pressure (Table 3). It is also possible that the calibration could be improved by including some additional model parameters, such as those relating to stretch-rate dependent force generation, even though they did not show a strong effect on single cell active tension upstroke (**Appendix Figure A1**). These additional parameters might have a stronger effect on the upstroke at ventricular scale than at single cell scale because the single cell simulations were performed under isometric conditions. The calibrated model parameters are listed in **Figure 1E**.

Validation by comparison to deformation and strain biomarkers, not used in calibration

Simulation showed broad agreement with the trend and magnitude of *in vivo* strain measurements from DENSE+cDTI images⁴⁷ for fibre strain, circumferential, radial, and longitudinal strain in mid-ventricular and four-chamber view slices. Simulated strain traces ($n=5004$ for short axis, $n=9282$ for long axis) showed higher variability than *in vivo* traces ($n=30$). Simulated fibre strain (Eff, **Figure 2A**) showed agreement in terms of transmural homogeneity when compared with *in vivo* measurements, and the magnitude of fibre shortening was similar to *in vivo* (compare with red line at -0.15 strain). Peak circumferential (Ecc), radial (Err), and longitudinal strains (Ell) in the simulations (**Figure 2B**) showed good agreement in magnitude with *in vivo* measurements (compare with red line at ± 0.15 strain). The timing of simulated peak strains occurred earlier than that *in vivo*. However, this is partially because the model was not calibrated to the cardiac cycle timings of the single individual in the DENSE+cDTI dataset.

In terms of deformations, the simulated systolic vs diastolic atrioventricular plane displacement of the calibrated biventricular (1.5 cm) showed good agreement with ranges reported in the literature (1.4 to 1.9 cm, **Table 1**).

When compared with literature reports of wall thickness changes, simulations showed thickening in systole in agreement with literature. The simulated end diastolic wall thickness (0.5 cm) was lower than the report range [0.57, 1.38] cm (**Table 1**, $n=4620$), and the end systolic wall thickness (0.7 cm) was also lower than the report range of [1.07, 1.75] cm (**Table 1**, 4620). The low wall thickness in both diastole and systole were consistent with the small heart size of the example used (96 g ventricular mass), and a larger and thicker heart would have greater wall thickening and fall better within the population ranges.

One-at-a-time sensitivity analysis

Figure 3 summarises how variability in single model parameters relate to simulated ECG (**Figure 3A**), pressure-volume (**Figure 3B**), displacement (**Figure 3C**), and strain (**Figure 3D**) biomarkers. Positive relationships between parameter and biomarker were shown in red and the negative were shown in blue.

ECG biomarkers (**Figure 3A**) were mostly affected by variabilities in ionic conductances (red block), with some small effects from the mechanical parameters (green block). QRS effects were very minimal, since conduction velocities were not altered in the analysis. QRS duration altered by SERCA (magnitude of change: 2.8 ms), G_{CaL} (1 ms), G_{Kr} (1.8 ms), and G_{NaL} (1.3 ms), while normalised QRS amplitude in lead V3 was very weakly affected by calcium sensitivity Ca_{50} (magnitude of change: 0.04), pericardial stiffness k_{epi} (0.01), and the passive stiffness parameter a_f (0.02) (**Appendix Figure A1.A**). The ST and T wave segments were much more sensitive to model

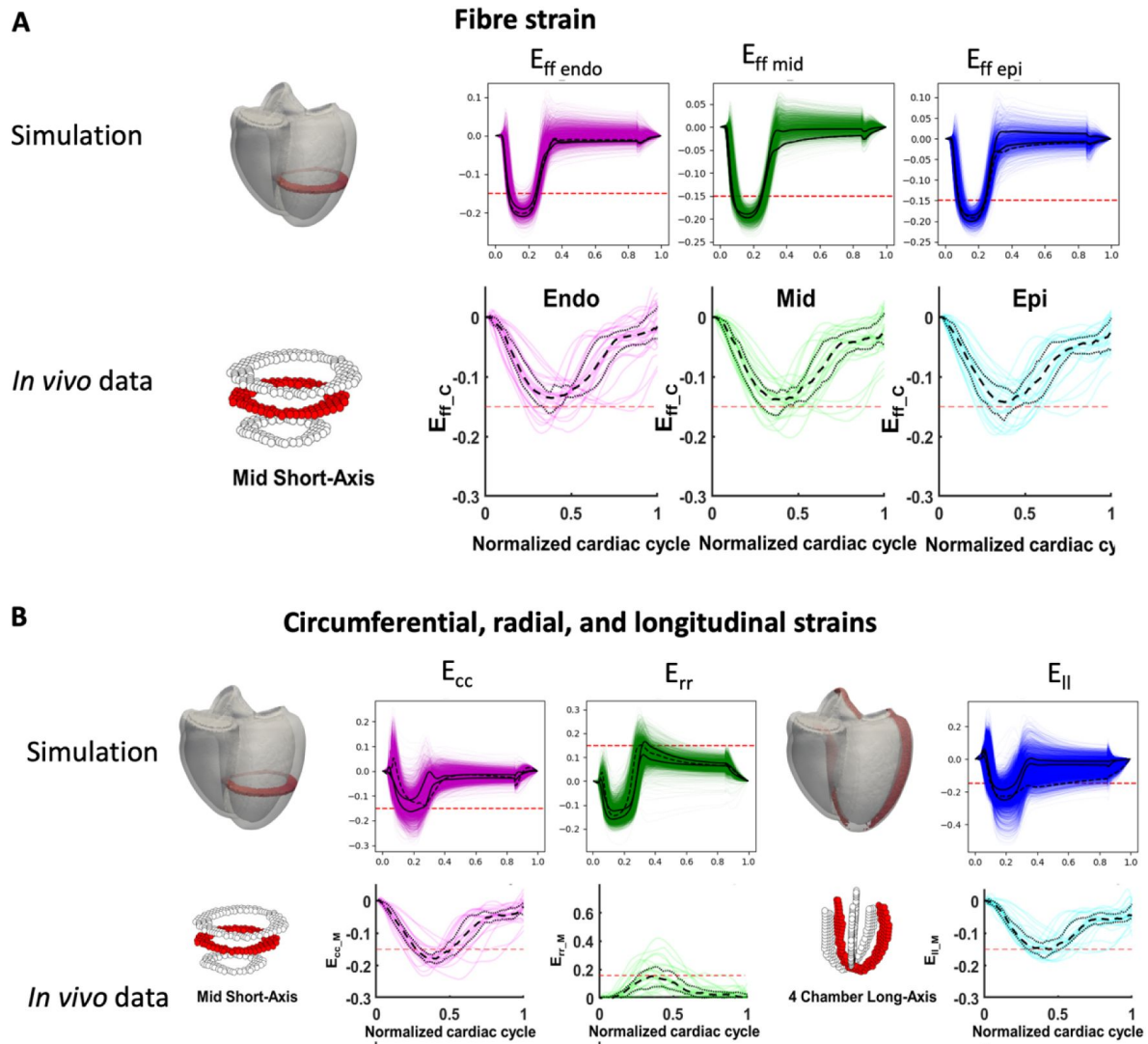


Figure 2

Comparison of simulated baseline left ventricular strains with non-invasive DENSE+cDTI MRI (in vivo) measurements.

(A) Comparison of strain along the myofiber direction measured in the mid-short-axis of the ventricles. (B) Comparison of strains in the circumferential and radial directions measured in a mid-short-axis slice and longitudinal strains measured in a four-chamber long axis slice.

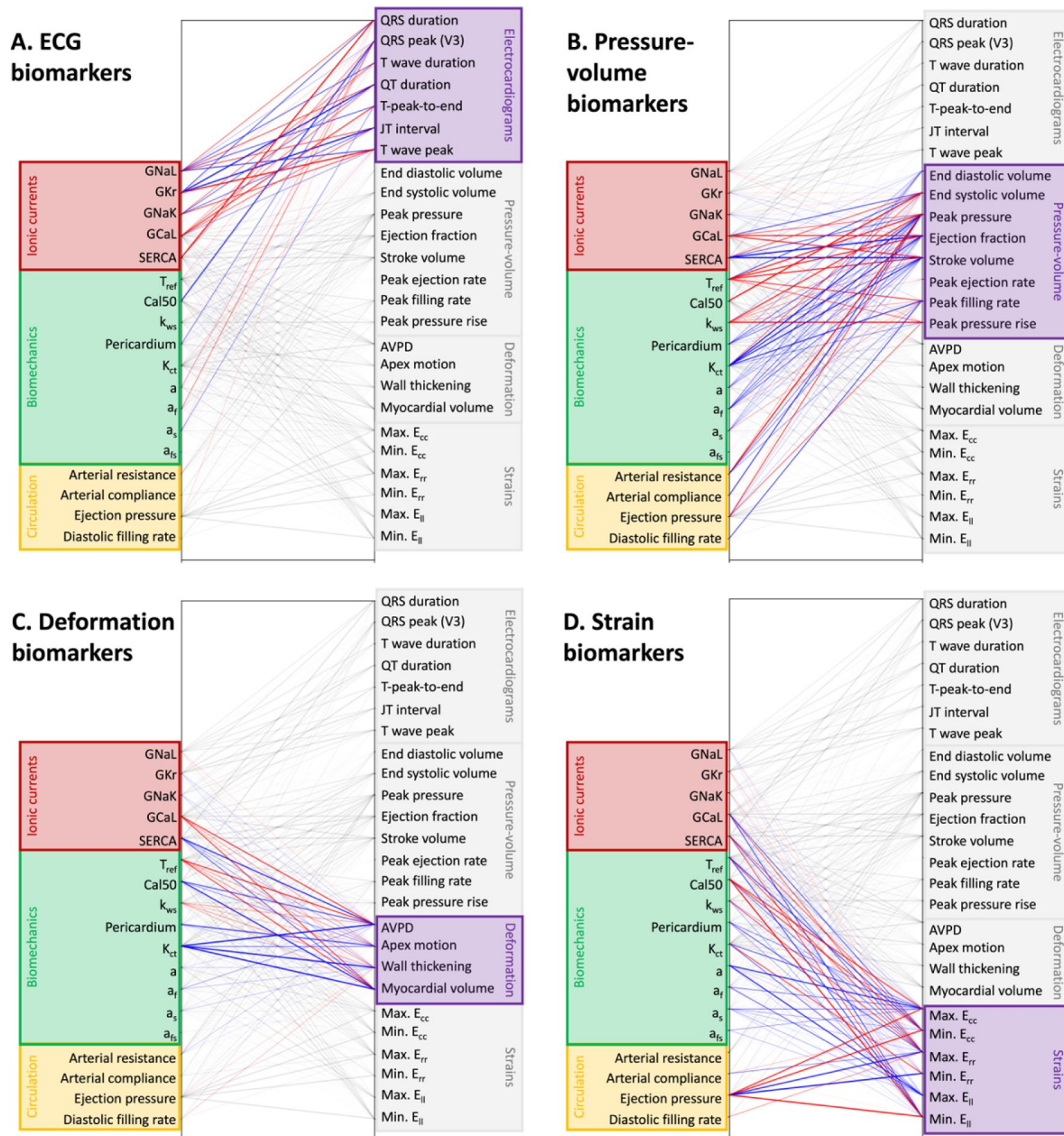


Figure 3

The effect of variability in single model parameters in cellular ionic conductances (red), passive and active biomechanical parameters (green), and circulatory model parameters (yellow) on simulated biomarkers of the ECG (A), pressure-volume (B), deformations (C), and strains (D).

Only relationships with p-value <0.05 were plotted. Positive correlations are shown in red, and negative correlations shown in blue. The thickness of each line indicates the normalised magnitude of effect and the transparency of each line indicates the r-value of the relationship.

parameters. The QT interval was strongly affected by GKr (magnitude of change: 173 ms), GNaL (92 ms), GNaK (72 ms), GCaL (69 ms), and SERCA (49 ms), with weak effects from calcium sensitivity (Cal50) (1.7 ms).

Normalised T wave amplitude was affected by GKr (magnitude of change: 0.1), GNaK (0.06), GNaL (0.06), and SERCA (0.04), GCaL (0.03), with very minor (<0.01) amplitude changes in select precordial leads for Tref (lead V4), Cal50 (lead V4), pericardial stiffness (lead V3) and Kct (lead V2) (**Appendix Figure A1.B** [↗](#)).

Pressure volume biomarkers (**Figure 3B** [↗](#)) were mostly affected by the mechanical (green block) and circulatory parameters (yellow block), alongside ionic currents (red block) that affect the calcium signaling system (GCaL and SERCA). **Appendix Figure A2** [↗](#) illustrates in detail the parameter effects on the pressure-volume loops. The ejection fraction was predominantly affected by cross-bridge power-stroke rate (k_{ws}) (magnitude of change in ejection fraction: 32 %), compressibility (Kct) (16 %), GCaL (15 %), peak active tension (Tref) (12 %), SERCA activity (10 %). There were also some effects on LVEF from GNaL (9%), arterial resistance (6 %), calcium sensitivity (Cal50) (8 %), and very weak effects from pericardial stiffness (4%), passive mechanical parameters a (3 %), a_f (2 %), a_s (4 %), and GNaK (2%), and GKr (2 %).

The peak systolic pressure was affected by the same parameters as the ejection fraction, except for GNaK, GNaL and GKr, and with the addition of arterial compliance (magnitude change: 1.1 kPa) and ejection pressure threshold (1.9 kPa). The peak rate of rise of ventricular pressure (dPdtmax) was strongly influenced by only the cross-bridge power-stroke rate (k_{ws}) (magnitude of change: 1 kPa/s), with weak effects from calcium sensitivity (Cal50) (0.6 kPa/s) (**Appendix Figure A3.A** [↗](#)). The peak ejection flow rate (PER) was influenced by Cal50, Tref, k_{ws} , while the peak filling rate (PFR) was influenced by a larger group of parameters including Tref, Cal50, Kct, GCaL, SERCA, and the bulk parameter controlling the diastolic filling rate (**Appendix Figure A3.B** [↗](#)).

Mechanical displacement biomarkers (**Figure 3C** [↗](#)) were predominantly affected by active mechanics (green block) and calcium system conductances (red block), with very minor effects from passive mechanical parameters. Systolic AVPD (**Appendix Figure A4.A** [↗](#)) was strongly influenced by Cal50 (magnitude of effect: 0.54 cm), GCaL (0.36 cm), Tref (0.33 cm), Kct (0.41 cm), SERCA (0.31 cm), with weaker effect from k_{ws} (0.15 cm), arterial resistance (0.09 cm), and pericardial stiffness (0.08 cm). Systolic wall thickness was predominantly affected by the same parameters as systolic AVPD, except for Kct, which had only a weak effect. Systolic percentage volume change (**Appendix Figure A4.C** [↗](#)) was strongly affected by the compressibility parameter (Kct) (magnitude of change: 12 %), as was expected, but also strongly affected by GCaL (9 %), Tref (8 %), SERCA (6 %), and k_{ws} (3 %). An increase in SERCA and Cal50 shifted the timing of peak volume reduction to earlier during systole.

Simulated strain biomarkers (**Figure 3D** [↗](#)) were predominantly affected by passive mechanical parameters (green block) and circulatory parameters (yellow block). The influence of parameters on the radial, circumferential and longitudinal strain patterns (**Appendix Figure A5** [↗](#)) followed broadly that of the AVPD and wall thickness. Of note was that at high values of the ejection pressure, the mid-ventricular circumference expands rather than contracts during systole and fails to return to the reference condition even after relaxation, pointing to failure of proper pumping function (**Appendix Figure A5.A** [↗](#)).

Multi-scale mechanisms underpinning ECG and LVEF relationship

Simulation results indicated that the link between ECG biomarkers and LVEF was chiefly underpinned by GCaL and SERCA, which had strong effects on both T wave biomarkers and the pressure-volume loop (**Figure 3** [↗](#)). Mechanical parameters T_{ref} , calcium sensitivity (Cal50),

compressibility (K_{ct}), and fibre stiffness (a_f) had much stronger effects on LVEF than on ECG signatures. On the other hand, ionic conductance parameters G_{NaL} , G_{Kr} , G_{NaK} had much stronger effects on the T wave than on LVEF.

Figure 4 illustrates the multi-scale mechanisms of G_{CaL} 's effect on both LVEF and ECG. A four-fold increase in G_{CaL} caused an eight-fold increase in cellular active tension peak and a 200 ms prolongation of active tension duration (**Figure 4A**), which resulted in a 15 mL decrease in end systolic volume and 2.5 kPa higher peak systolic pressure at the ventricular scale (**Figure 4C**).

G_{CaL} 's dramatic effect on end systolic volume was facilitated by 10 % higher systolic myocardial volume compression, 4.5 % higher systolic longitudinal contraction, 3.5 % higher systolic circumferential contraction, and 0.4 cm lower (towards apex) systolic position of the atrioventricular plane (**Figure 4B**). The increase in G_{CaL} caused an increase of 60 ms in cellular action potential duration (**Figure 4A**), which prolonged the global repolarisation time (**Figure 4D**) and led to the prolongation of the QT interval (**Figure 4C**). G_{CaL} did not have a significant effect on the activation pattern and conduction velocity in our simulations (**Figure 4D**) or on the QRS complex of the ECG (not shown).

Figure 5 illustrates how a four-fold increase in SERCA activity caused a 12% decrease in LVEF, with non-monotonic effects on the QT interval. The increase in SERCA conductance increased diastolic residual active tension (**Figure 5A**), which was sufficient to cause 3 mL reduction in end diastolic volume (**Figure 5C**). Even though increasing SERCA increased peak active tension at the cellular scale by 9 kPa (**Figure 5A**), it reduced peak systolic pressure and LVEF at the ventricular scale (**Figure 5C**). This was because increasing SERCA hastened the arrival of peak active tension by 50 ms and reduced its duration by 100 ms (**Figure 5A**), which meant that the mechanical contractions were not given enough time to developed, resulting in earlier and weaker systolic contractions (**Figure 5B**). On the electrophysiological side, while SERCA did not elicit strong APD changes at the single cell simulation, when combined with non-isometric conditions in ventricular simulations, it caused action potential prolongation at values higher than 1.5-fold and resulted in prolonged repolarization (**Figure 5E**).

Discussion

In this study we presented the calibration, validation and sensitivity analysis of human healthy ventricular electromechanical modelling and simulation, following the ASME V&V40-based assessment guidelines. The first step was compiling biomarkers values from multi-modal data characterizing pressure-volume, ECG, displacement, and strain behavior of human healthy ventricles, and separated the data for calibration and validation. The calibration was informed by unravelling the interlinking relationships between simulated biomarkers and model parameters, especially highlighting the plethora of model parameters that affect the ejection fraction. We demonstrated multi-scale mechanistic explanations of the relationship between LVEF and ECG underpinned by variability in L-type calcium channel conductance and SERCA activity.

Previous examples of applying the ASME V&V40 framework for cardiac model credibility assessment have focused primarily on evaluating computational models of medical devices such as the left ventricular assist device¹⁸ and artificial heart valves¹⁹, where model validation was performed by comparing to experimental recordings specifically designed for this purpose. These studies, as well as other examples of model validation without explicit reference to the ASME Standard, commonly have narrow focuses on specific diseases and/or therapies, and the biomarkers used for model evaluation were similarly limited in scope. In the case of disease models, calibrations were commonly performed at the cellular or tissue scale and validation was performed at the ventricular scale through comparisons with known ECG or ejection fraction phenotypes^{9,12}. While these studies have been valuable for providing model credibility in

GCaL

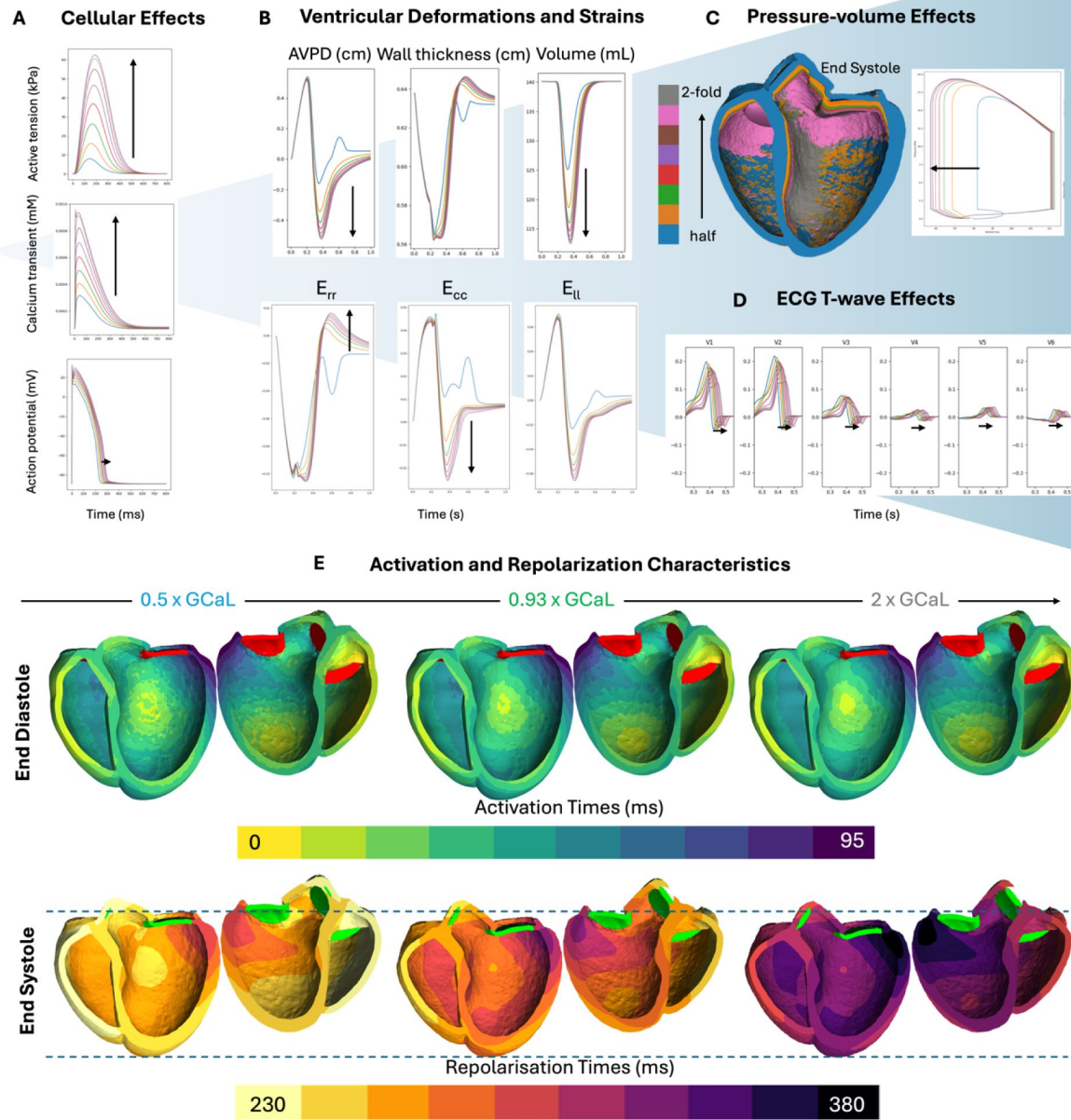


Figure 4

Multi-scale effect of GCaL on active tension, action potential duration (A), ventricular deformation and strains (B), the pressure-volume loop (C) and precordial ECG leads (D).

ECG characteristics were explained by the activation and repolarization maps (E).

SERCA

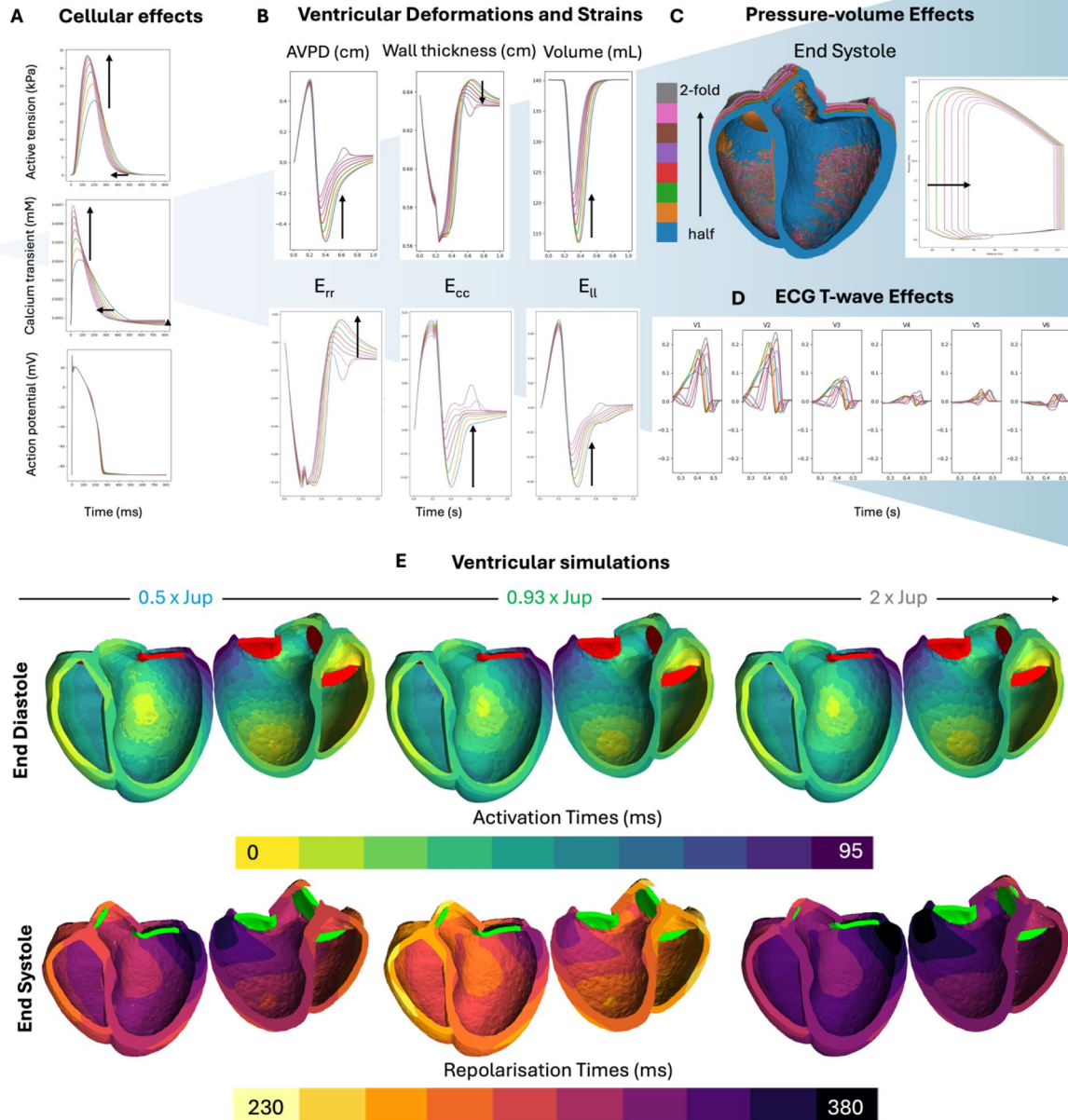


Figure 5

Multi-scale effect of SERCA conductance on active tension, calcium transient, action potential duration (A), ventricular deformation and strains (B), the pressure-volume loop (C) and precordial ECG leads (D).

ECG characteristics were explained by the activation and repolarization maps (E).

their specific disease/therapeutic contexts of use, they have not been designed to give credibility to the underlying computational electromechanics framework. In this study, we instead focused on building a comprehensive calibration and validation strategy for baseline healthy biventricular electromechanics. Through considering biomarkers at multiple scales and across both electrophysiology and mechanics, we hoped to establish a firm foundation of credibility for computational electromechanics that can be refined in future studies.

We showed in this study that literature values for model parameters do not automatically provide good baseline simulations. We proposed a sequential calibration strategy based on knowledge of model sensitivities, as detailed in the methods. This approach can be adapted for more sophisticated numerical techniques involving the use of emulators and Bayesian statistics to enable faster and more comprehensive explorations of the parameter space^{24,26,86}.

Our sensitivity analysis results showed that in the simulated ECGs, the T wave was far more sensitive to electrophysiological parameters than to mechanical ones, within the ranges of uncertainty expected from literature reports of each parameter. In our simulations, the ionic conductances GNaK and GNaL primarily affected the T wave through altering the action potential duration (**Appendix Figure A1**) while having minimal effect on mechanical deformation or strain. GCaL and SERCA strongly affected the repolarization pattern and deformation and strain patterns during systole (**Figures 4** and **5**). However, other parameters that had a strong effect on deformation and strain, such as Tref, Cal50, pericardial stiffness and Kct had only a minor effect on the T wave (**Appendix Figure A2**). Previous studies on this question focused on comparing either end diastolic versus end systolic geometry⁸ or a static versus dynamic geometry⁸⁷ when simulating the T wave. These simulations involved a much larger perturbation in deformation than relevant to physiological situations and showed a larger effect on the T wave amplitude than our simulations. Within the context of physiological systolic deformations, however, our results showed that deformation uncertainties had only minor effects on T wave amplitude.

Our simulations showed only minor changes to QRS amplitudes in response to variation in those parameters that influenced the end diastolic geometry: calcium sensitivity, through altering diastolic residual active tension, pericardial stiffness, through restricting diastolic inflation, and the passive mechanical parameters a and af , through altering bulk stiffness and stiffness along the myofibre direction. However, it is possible that the magnitude of changes to the QRS was underestimated in our simulations, because the local activation times on the endocardial surface in our simulations were fixed. This meant that our simulations did not explore the effect of variations in end diastolic geometry on the conduction velocity of the fast endocardial activation layer or on Purkinje conductivity. It should also be noted that other parameters that affect the activation pattern, such as GNa and conduction velocities, were not included in this analysis.

The fact that the initialised literature values of model parameters failed to achieve LVEF above 50 % pointed to limitations in using ex vivo tissue measurements to represent in vivo function. In our simulations, we saw that the LVEF was strongly sensitive to changes in the incompressibility of the tissue (Kct), such that an increase in compressibility of the myocardial tissue helped to increase LVEF. The systolic volume has been reported to change up to 13%⁸¹, pointing to limitations of assumptions of incompressibility in the modelling of the myocardial tissue. Additionally, our model did not achieve a high amount of wall thickening in validation, which meant that the high compressibility required in our model to achieve LVEF could be compensating for a lack of sufficient wall thickening, in addition to anatomical effects. Previous experimental and simulations studies have shown how wall thickening is related to sliding of sheetlets^{88,89}. However, in our simulations we did not see a significant effect of the shear stiffness parameter a_{fs} on wall thickness or LVEF. One possible avenue of exploration would be the effect of sheet and fibre orientations on the thickening effect, as explored in a previous simulation study⁸⁹.

Displacement and strain biomarkers (**Figure 2D** [↗](#)) were far more sensitive to calcium handling dynamics, cross-bridge cycling rate, and contractility than to passive stiffness parameters, which indicates they would be better indicators of systolic rather than diastolic dysfunctions in HF. The sensitivity of strains and displacements to the pericardial stiffness parameter was unsurprising due to the association of reduced strains in diseases affecting the pericardial sack⁹⁰ [↗](#). The sensitivity of longitudinal and radial strain to the incompressibility of the myocardium (Kct) mirrors the sensitivity of the LVEF to that parameter, which was in accordance with the fact that those two strains were surrogate markers of LVEF and show better predictive power⁸³ [↗](#),⁸⁴ [↗](#).

Conclusions

In this study, we show the importance of applying vigorous VVUQ evaluations of ventricular electromechanics for achieving physiological simulations. We set the basis for a strategy for calibrating and validating baseline electromechanical modelling and simulation frameworks for cardiac Digital Twins. We compiled a comprehensive list of biomarkers for evaluating healthy electromechanical function, and grouped the dataset into non-overlapping calibration and validation sets. We provide evidence of credibility through calibration to haemodynamic and ECG biomarkers and validation by comparison to strain and deformation biomarkers and uncertainty quantification. Furthermore, our analyses highlighted the interplay between cellular, tissue, and haemodynamic parameters on the LVEF and provided multi-scale explanations of its link with ECG biomarkers. Taken together, the study paves the way towards credible electromechanical cardiac Digital Twins.

Appendix

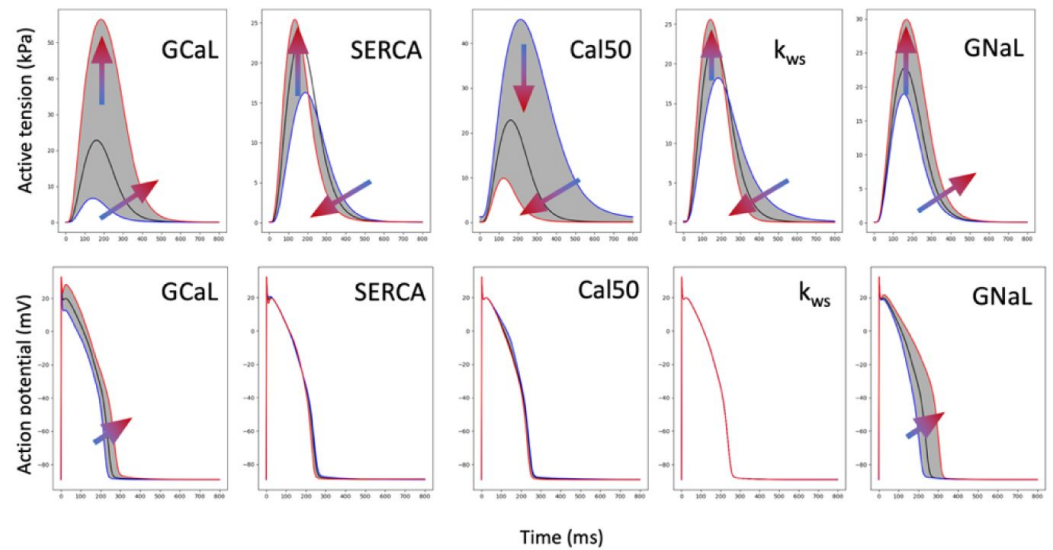
1. Pressure volume calibration strategy design

The goal of the calibration of the initialised set of parameters was to increase LVEF, increase systolic pressure, increase peak filling rate, but decrease peak ejection rate and lower dP/dt_{max} . The LVEF and peak systolic pressure had the highest degree of importance for calibration, since they were implicated in a variety of diseases and was well established. The filling rates and dP/dt_{max} took a secondary role, since the data came from much smaller sample sizes, and the filling rates were measured using echocardiography techniques, which were not as reliable as volume measurements in CMR (**Table 1** [↗](#)).

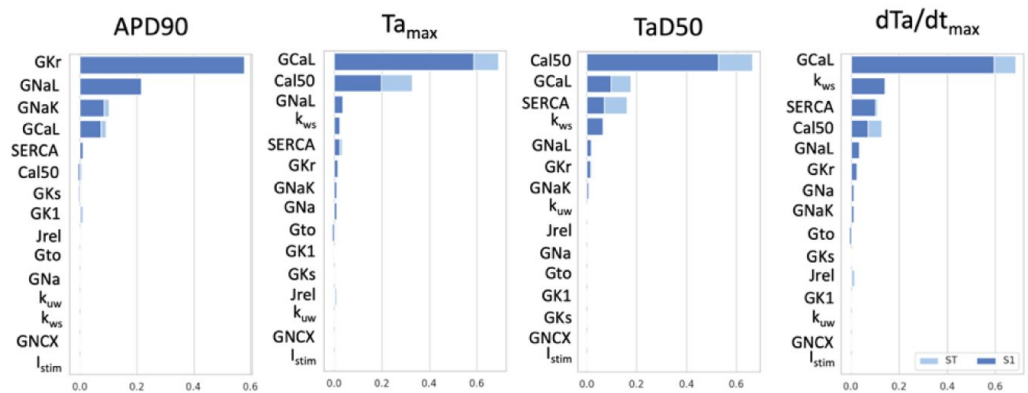
From the uncertainty quantification, we saw that a handful of parameters can increase both LVEF and peak systolic pressure simultaneously: Tref, k_{ws} , GCaL, SERCA, Cal50, Kct (**Figure 3** [↗](#)). However, Tref and k_{ws} effects saturate at high values, large changes in SERCA and GCaL can become arrhythmic substrates, Cal50 has a non-monotonic effect on LVEF due to its effect on the diastolic function, and dramatic reductions in Kct brings unrealistic amounts of systolic volume change. A combination of changes in these parameters was needed to dramatically improve LVEF and peak systolic pressure, and in practice, was insufficient by themselves. Decreasing arterial resistance was also needed to help increase LVEF. However, this comes at the cost of reducing peak systolic pressure, which needed to be counterbalanced by increasing the ejection pressure threshold. An increase in ejection pressure increased time spent in isovolumic contraction and isometric force development, and thereby increased the peak systolic pressure. An additional challenge was that there was not a single parameter in our analysis that could decrease peak ejection flow rate (Tref, k_{ws} , Kct) and dP/dt_{max} (k_{ws}) without also decreasing LVEF at the same time. This meant that we needed to achieve higher than healthy LVEF first. With all this in mind, the calibration strategy was as detailed in the Methods section.

2. Cellular global sensitivity analysis

A. Key parameters for active tension dynamics



B. Sobol indices of sensitivity analysis



C. Scatter plots for key parameters

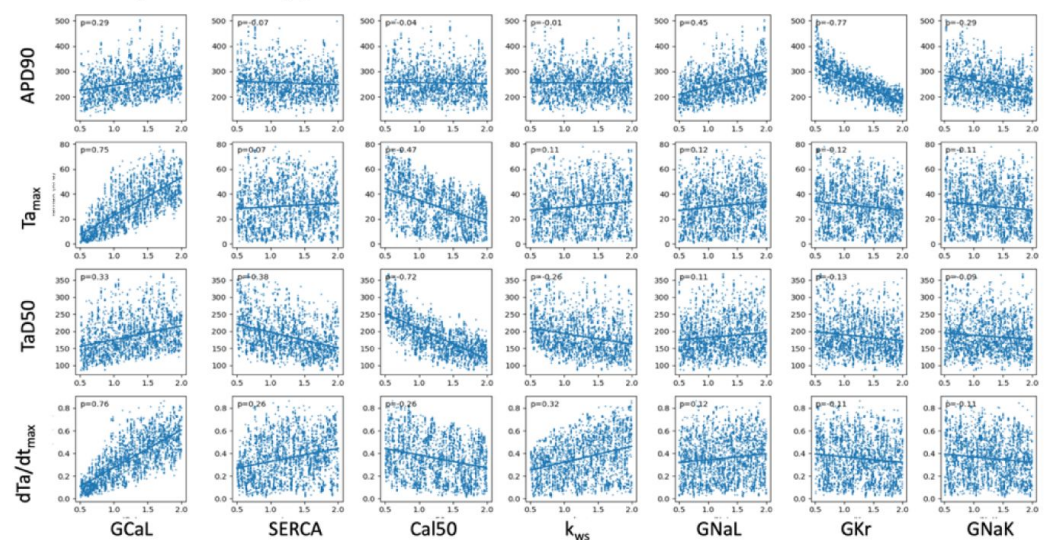


Figure A1

Global sensitivity analysis (C) using cellular model of ventricular electromechanics show the effect of the top five model parameters on active tension (A) and action potential duration (A and B) biomarkers.

3. Additional ventricular sensitivity analysis results

Left ventricular pressure volume loops

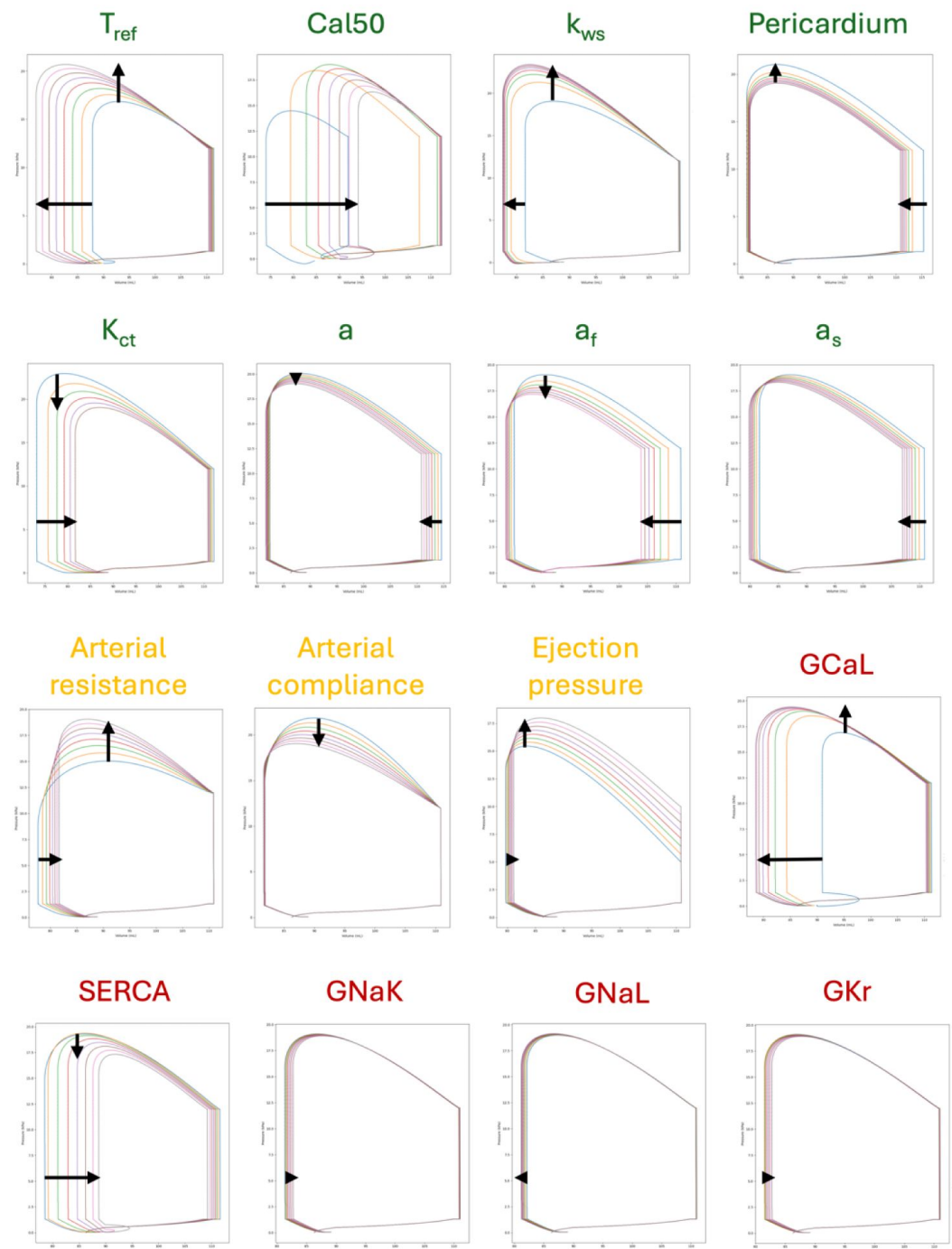


Figure A2

Uncertainties in simulated pressure volume dynamics were influenced by uncertainties in mechanical (green), circulatory (yellow) and ionic conductance (red) parameters of the model.

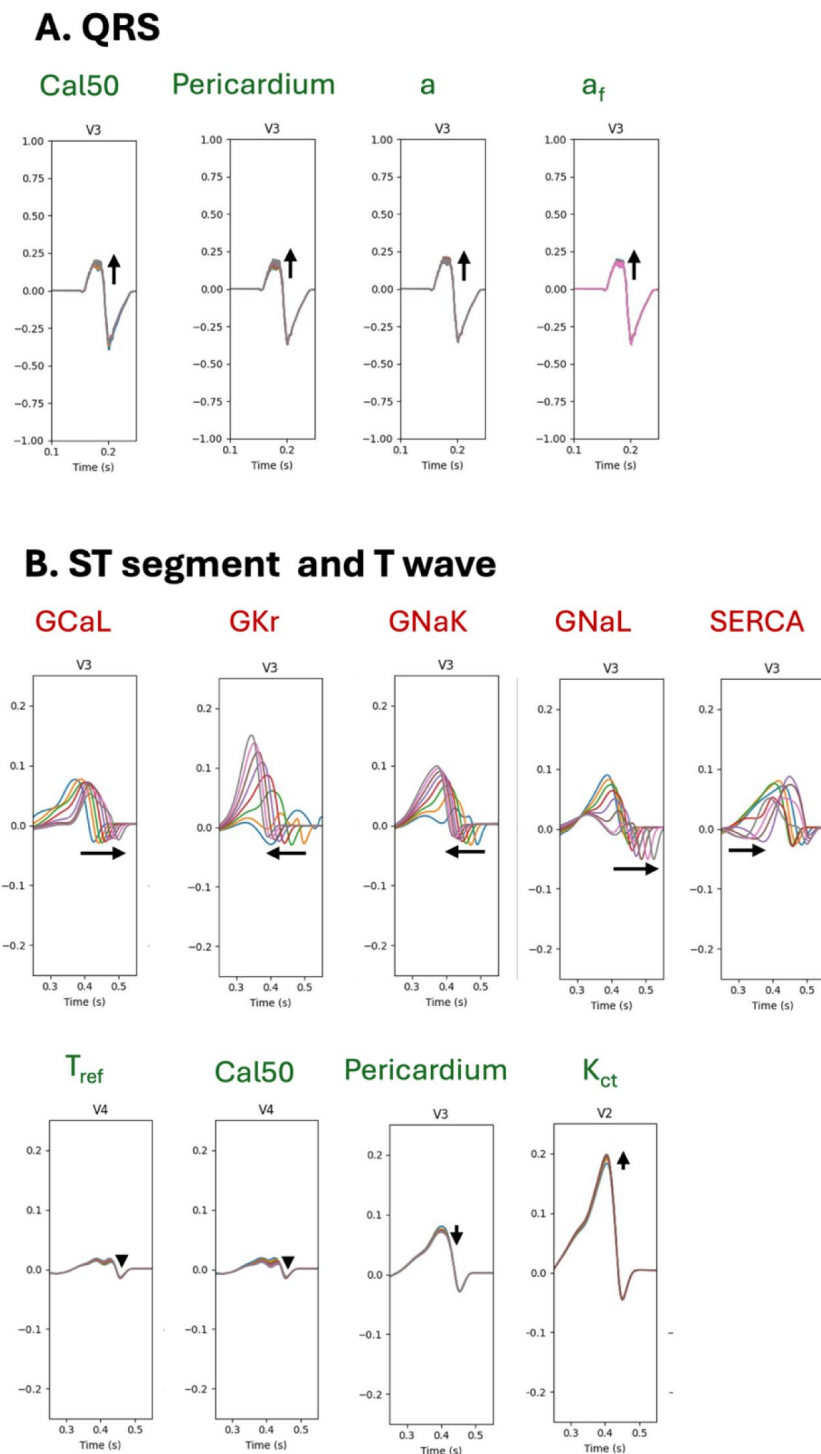
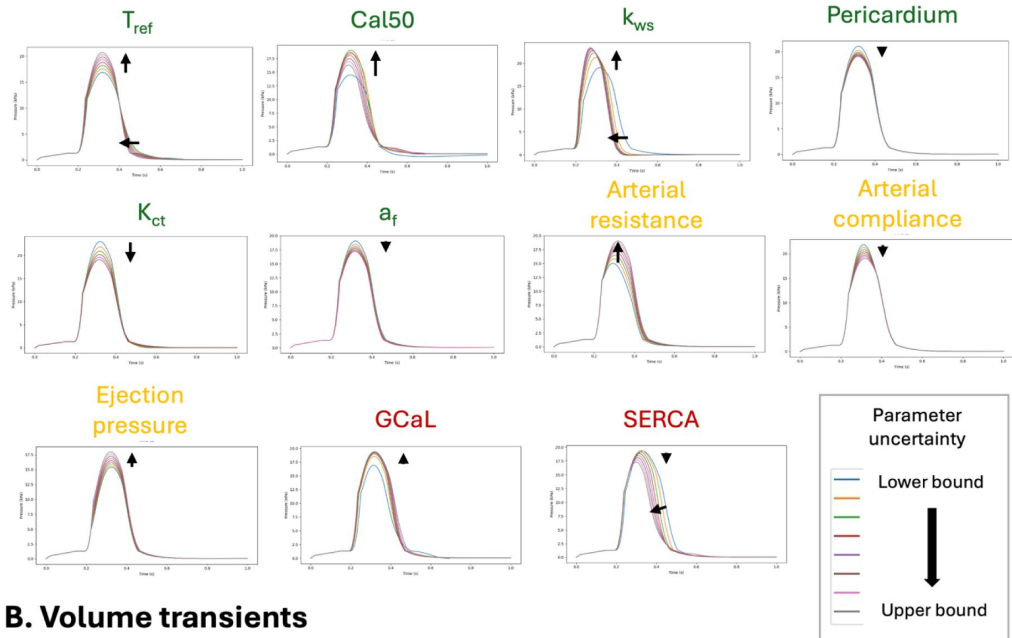


Figure A3

Of the parameters included in the analysis, the QRS section of the ECG showed minor sensitivity to uncertainty in

mechanical parameters (green labels) (A) while the ST and T wave segments of the ECG were strongly sensitivity to uncertainties in ionic conductances (red labels), and only showed minor sensitivity to some mechanical parameters (green labels) (B).

A. Pressure transients



B. Volume transients

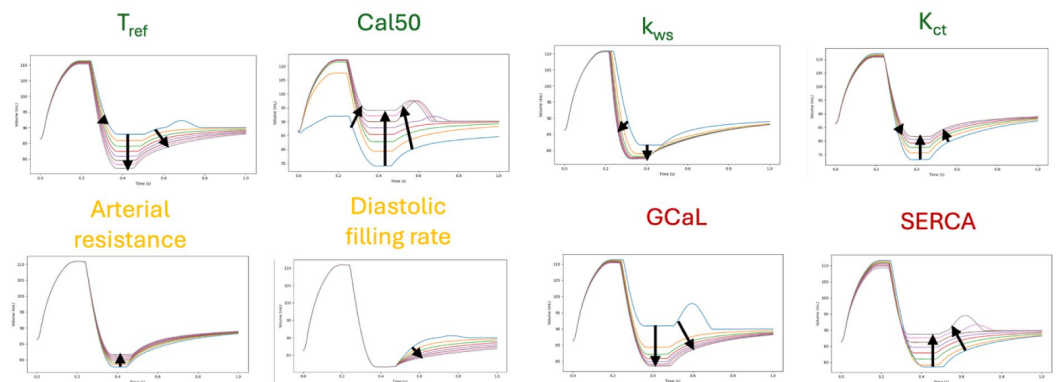
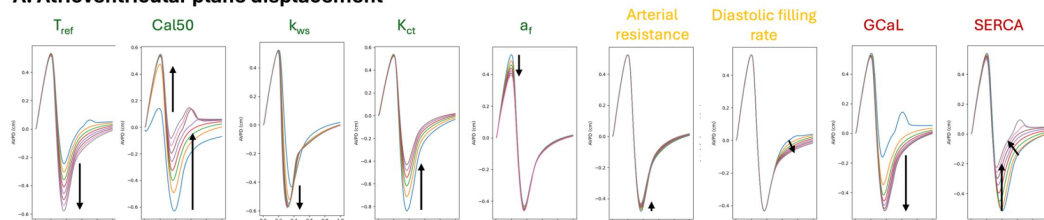


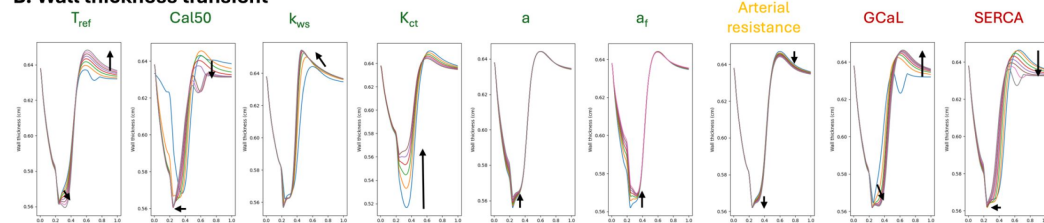
Figure A4

Pressure (A) and volume (B) transients in response to parameter uncertainties in mechanical properties(green), circulation (yellow), ionic conductances (red).

A. Atrioventricular plane displacement



B. Wall thickness transient



C. Myocardial volume

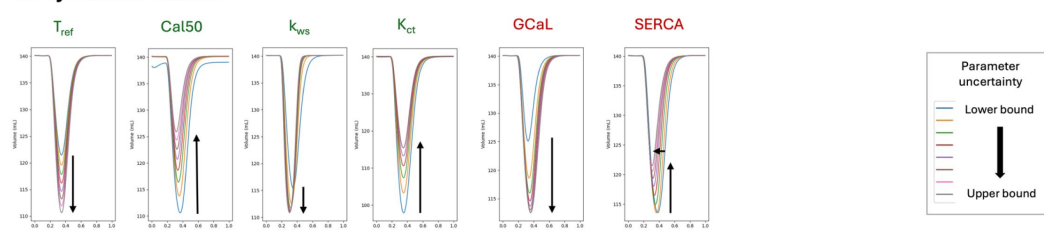
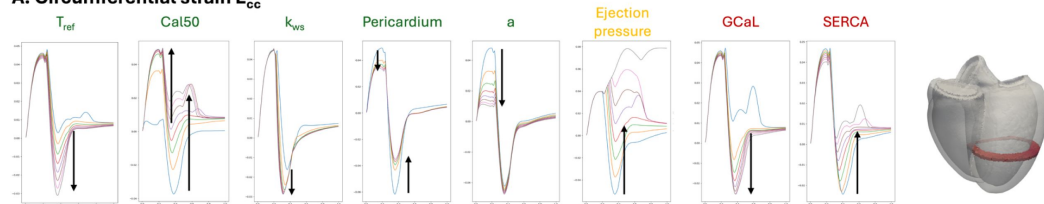


Figure A5

Key aspects of the ventricular deformation, including atrioventricular plane displacement, wall thickness changes, and myocardium volume changes were affected predominantly by mechanical parameters (green) and ionic conductance parameters (red), with weaker effects from circulatory parameters (yellow).

A. Circumferential strain E_{cc}



B. Longitudinal strain E_{ll}

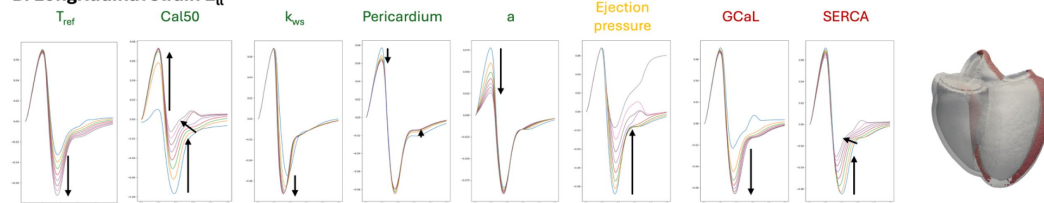


Figure A6

Uncertainty in simulated strain transients were influenced by uncertainties in mechanical (green), circulatory (yellow), and ionic conductance (red) parameters.

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Reviewer #1 (Public review):

Summary:

The study by Wang et al. investigates cardiac electromechanical modeling and simulation techniques, focusing on the calibration and validation of ventricular models according to ASME V&V40 standards. The researchers aim to calibrate model parameters to align with key biomarkers such as QRS duration and left ventricular ejection fraction, and validate the model against independent measurements such as displacement and strain metrics. The authors also examine the impact of parameter variations on deformation, ejection fraction, strains, and other biomarkers. The overarching aim of the study is to give "credibility to the underlying computational electromechanics framework" and to "pave the way towards credible cardiacelectromechanical Digital Twins."

Strengths:

- (1) The study presents a solid validation strategy for cardiac models based on independent data.
- (2) It integrates electrophysiological, mechanical, and hemodynamic biomarkers for sensitivity analysis and calibration.

Weaknesses and Limitations:

- (1) Model Assumptions: The study employs simplified modeling assumptions that are not state-of-the-art, e.g.,
 - a) Isotropic scaling of the mesh to generate an unloaded reference geometry.
 - b) Simple afterload and preload models that fail to produce physiological results.
 - c) Simplified epicardial boundary conditions.
- (2) Numerical Framework:
 - a) The mesh resolution and/or the numerical framework used for the mechanical part appears to suffer from known numerical artifacts (locking effects), leading to overly stiff or inaccurate behavior in finite element analysis. This results in an artificially stiff response to deformation, which is compensated by setting active contraction to ten times the value

reported in the literature. The authors attribute this to limitations in using ex vivo tissue measurements to represent in vivo function, although similar issues were not observed in previous works.

b) Further, the authors employ the monodomain model for the simulation of the electrical excitation and relaxation on a relatively coarse grid with an approximate edge length of 1mm. This resolution is known to be insufficient for reliable results in organ-scale electrophysiology modeling.

(3) Geometrical model and digital twin: The geometrical model, taken from a public cohort and calibrated to an ECG of another individual along with population-averaged values from a databank (UK Biobank), and unrelated measurements from surgical procedures, can hardly be considered a digital twin. Further, validation of the model was then performed against data from yet another cohort.

(4) Calibration procedure: There are apparent flaws in the calibration procedure, or it is not described in sufficient detail. The authors dedicate significant effort to motivating parameter ranges, but in the end they use mostly other parameters for the calibration process, aiming to maximize left ventricular ejection fraction. It is not clear whether the chosen parameters result in, e.g., physiological calcium traces or calibrated parameters that are within physiological ranges.

(5) Goodness of fits, e.g., a direct comparison of the measured and the simulated ECG, are not provided to assess calibration quality.

(6) Due to these limitations and weaknesses, the authors fall short of achieving some of their goals, particularly establishing credibility for the underlying computational framework and in reproducing healthy pressure-volume loops, and in achieving physiological simulations while using physiological or reported ranges for the calibrated parameters.

For example, a key physiological requirement is that the right and left ventricular stroke volumes are approximately equal in a heart beating at a limit cycle, as the blood pumped by the right ventricle into the pulmonary circulation must match the amount pumped by the left ventricle into the systemic circulation. This balance is not achieved in this study.

(7) The conclusive claim that "the study paves the way towards credible electromechanical cardiac Digital Twins" is not supported. The model exhibits non-physiological behavior, requires unsupported parameter alterations (such as a 10-fold active stress scaling), and does not represent a digital twin, as model data are drawn from various unrelated, non-patient-specific sources.

Conclusion:

Overall, this reviewer considers that the study requires a major revision, including improvements in numerical methods, modeling choices, and checks for physiological behavior. Nevertheless, the provided tables with averaged values from the UK Biobank and the presented validation strategy could be valuable to the research community.

<https://doi.org/10.7554/eLife.106555.1.sa1>

Reviewer #2 (Public review):

The authors present an interesting study on calibrating and validating a biventricular cardiac electromechanical model. This is an important contribution, but some questions remain about the quantitative validation and verification aspects of the study.

Major comments:

(1) The title and paper stress the importance of validation on several occasions. However, the actual validation performed is limited to the section in lines 427-439. Furthermore, it is entirely qualitative, making assessing the model's quality difficult. Most of the paper is focused on sensitivity analysis, which is also interesting but unrelated to validation. Can you include a quantitative comparison with deformation biomarkers? E.g., spatially quantify strain differences between simulation and in vivo data, or overlay the current configuration of the geometry with MRI in various views, and calculate a displacement error norm.

(2) You mention the ASME V&V40 standards throughout your paper. Yet, you only address the "second V" validation, ignoring the "first V" verification. How did you ensure that your computational models are implemented correctly?

(3) All parameters discussed in this publication are physical parameters. What is the sensitivity of your model outputs concerning computational parameters?

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